Survey interviews across three waves

Two independent NORC extracts support six descriptive questions about US adults. The usable sample depends on which questions each interview contained.

Why these years

A includes 2014 and 2016. B includes 2016 and 2018. These consecutive waves give roughly 5,000 interviews per extract, retain coverage of every requested concept and supply a shared wave for testing duplicate handling. The same years serve all questions. Each query defines its own eligible population.

Wave	Cut A	Cut B	Distinct interviews
2014	2,538	-	2,538
2016	2,867	2,867	2,867
2018	-	2,348	2,348
All waves	5,405	5,215	7,753

All 2,867 shared interviews agree field for field. They comprise 53.04% of A and 54.98% of B. The 2012-2016 alternative gave fewer observations for web duration. The 2016-2021 alternative introduced survey mode and wording changes in 2021. Each would still meet the overlap requirement.

What the files contain

Explorer supplied 32 fields and an embedded codebook. Each archive contained a strict JSON file with aligned column arrays and another file formatted as a Python literal. Both representations produced identical records. The repository preserves the strict JSON bytes and records their hashes.

Usable answers / interviews	2014	2016	2018
EVWORK / past employment	887	1,023	816
SATJOB / work satisfaction	1,895	1,779	1,739
TVHOURS / daily TV time	1,669	1,883	1,555
INTTECH / technology interest	1,238	1,386	1,173
Both web time components	1,398	1,587	1,362

The pre-build profile in docs/profile.csv covers every variable and year. Ballot C contributed zero valid NEWS, TVHOURS, COMPUSE or WWWHR answers in all three waves. TV and news cover 65.7%-66.4% of interviews; web hours cover 55.1%-58.0%, with additional routing inside A/B. Q4 and Q6 therefore describe responding subsets. See 09_ballot_coverage.csv.

What each answer means

All concepts have observations in 2014, 2016 and 2018. Having a field in a wave does not mean every respondent received that question.

Question	Fields and interpretation
Working now	WRKSTAT. Full time, part time and temporarily absent from a job count as working. Unknown status remains unknown.
Worked a year	EVWORK asks about working as long as one year. It is routed to retired people, students, homemakers and the other nonworking category. Duration for current workers and job seekers cannot be recovered.
Education	DEGREE supplies comparable credentials for industry groups. EDUC measures years of schooling.
Industry	INDUS10 identifies the industry. Restrict to working respondents because nonworkers can report an earlier industry.
Work satisfaction	SATJOB asks about work. Restrict the TV comparison to workers. In 2016 ballot A, form Y uses another wording; keep it outside the standard item.
Afterlife	POSTLIFE separates yes/no belief from uncertainty, no answer and unavailability.
Prayer	PRAY is an ordered frequency. Report the distribution within upbringing groups.
Raised religion	RELIG16 measures upbringing. RELIG is current preference; ATTEND and RELITEN describe attendance and affiliation strength.
Newspaper	NEWS measures reading frequency. Compare daily readership as a share of valid answers on ballots A and B.
Television	TVHOURS measures hours per day on ballots A and B. Zero hours contributes zero to the mean.
Technology	INTTECH measures interest in new inventions and technology issues. Its observed ballot/form coverage changes by year.

Media routing and other fields

WWWHR plus WWWMIN / 60 measures weekly non-email web time only when both components are valid. COMPUSE, WEBMOB, EMAILHR, EMAILMIN and USEWWW describe routing. A skipped USEWWW does not establish zero use. AGE, SEX, REGION, FORM, BALLOT, MODE and two person weights preserve context.

Valid INTTECH appears on B/X, B/Y and C/Y in 2014 and on A/X, A/Y and B/X in 2016 and 2018. Of 3,797 valid answers (48.97% coverage), 41.30% were very interested, 45.96% moderately interested and 12.75% not at all. See 08_technology_interest.csv for each year.

What the seven queries show

Percentages use observed, eligible responses. Each substantive query reads the respondent table, whose (year, ID) grain is tested against the input membership.

1 Prayer by religion raised in

7,659 interviews have both responses. Several-times-a-day prayer was reported by 1,413/4,007 raised Protestant (35.26%), 601/2,515 raised Catholic (23.90%) and 28/49 raised Muslim (57.14%). Small groups are particularly sensitive to a few responses. The full frequency distribution and denominators are in 01_prayer.csv.

2 Belief in life after death

Yes responses were 1,783/2,242 (79.53%) in 2014, 2,089/2,588 (80.72%) in 2016 and 1,720/2,123 (81.02%) in 2018. Each wave had zero recorded not-asked cases. Excluded uncertain/no-answer counts were 296, 279 and 225 respectively.

3 Most frequent industry among workers

Construction led for less than high school (53/384) and high school (186/2,275). Hospitals led for associate degrees (41/398). Elementary and secondary schools led for bachelor's (109/1,037) and graduate degrees (94/582). These are pluralities among workers with known credentials and industry.

4 TV hours by job satisfaction

Satisfaction	Very satisfied	Moderately satisfied	A little dissatisfied	Very dissatisfied
Group size	1,415	1,091	251	88
Mean TV hours/day	2.32	2.37	2.39	2.55

Of 4,712 current workers, 4,423 answered SATJOB. Of those, 3,911 (88.42%) were very or moderately satisfied. Among the 2,845 who also reported TV time, 2,506 (88.08%) were satisfied. Median TV time was 2 hours in every group. Neither test detected an association (Spearman rho = -0.011, higher rank = more satisfied, p = 0.570; Kruskal-Wallis p = 0.444). These unweighted sample comparisons assume independence and omit survey clustering.

5 Ever worked at least one year

Of 3,034 nonworkers, 306 looking for work were routed out and two eligible respondents had no usable answer. Of the 2,726 who answered, 2,467 said yes (90.50%). Assigning all 308 unknowns to no or yes gives bounds of 81.31%-91.46% for all nonworkers.

6 Changes between waves

Daily readership was 24.90%, 21.11% and 20.01%. The first drop was -3.78 points (2.67 approximate SEs); the next was -1.10 points (0.80 SE). Mean TV hours were 2.98, 3.03 and 2.94; both changes were under one SE. These unweighted independent-wave comparisons do not establish a national trend.

7 Overlap audit

The 7,753 analysis rows have 7,753 distinct (year, ID) keys but just 2,867 distinct IDs. Joining or deduplicating on ID alone would lose 4,886 interviews. The grain test and 07_overlap.csv verify the common input to Q1-Q6.

Operation, cost and oversight

Python validates and loads files. DuckDB holds raw rows and analytical tables in one local database. dbt makes the transformations, definitions and tests inspectable.

From source to respondent

A source row belongs to one file and ordinal. A respondent key is the survey year plus the within-year ID. Membership stores extract keys. Materializing answers decodes each label once per build. Materializing respondents avoids repeating the pivot for each analysis. Lower snake case is used throughout; _value is numeric, _status explains availability and _n is a count. Versioned seeds describe each variable, its domain, reviewed years, source codes and missing types. Genuine zero, no answer, uncertainty, inapplicability, absent property and year-unavailable responses remain distinguishable. Original labels stay accessible.

Reruns and failure

Every run revalidates inputs against the current codebook. File hashes skip unchanged inserts. Conflicting overlap or malformed input rolls back the load. New files enter raw storage once. The next dbt build regenerates the small downstream tables. Missing or renamed inputs stop the run. Results are staged until all queries pass; obsolete CSVs are removed before the success marker is written. The run log records hashes, counts, stages and exceptions. Local tests compared actual A+B and B+A, repeated each load, and checked rollback on a deliberate conflict.

Measured change and scale

The initial membership view parsed keys from JSON on every audit. Screening all eleven SQL queries identified the overlap audit as slowest. Materializing the four membership columns reduced the median from 29.80 to 5.80 ms on this machine, single thread, one warm-up and three timed runs. Both versions returned the same result. The key-table build took 0.21 s on the clean run, including dbt overhead. The plans and timings are in query-plans.txt.

The 30 non-key variables produce 232,590 answer rows today. At 75,700 interviews and 7,000 variables the dense expansion approaches 530 million rows. Full rebuilding, JSON parsing and memory for whole input arrays would become the first constraints. Project only needed variables, store typed columnar partitions, and rebuild affected waves when measured costs warrant it. Keep versioned semantics per source. A local runner and pinned virtual environment suffice for these manual batches. PostgreSQL and Airflow would add two services to operate for a job that runs on demand.

Contribution, limitations and next checks

I set the scope and take responsibility for the conclusions I present. I asked for the choice of years to be justified against ballot coverage, eligibility and overlap, approved the three-wave plan, and supplied both Explorer archives. I kept the build local and challenged the TV, work-history and wave-change interpretations. Codex wrote the pipeline and SQL, ran the checks and drafted the report. A second AI reviewer checked the bonus scope and challenged equating web use with access. My review prompted the separate counts for routed-out and unanswered cases and the statistical checks reported here.

The recorded initial build interval was about ten minutes from the design commit to the first verified report. This measures only that stage; the chat does not establish my total personal hours across planning, extracts and review. With more time I would assess survey-design-adjusted uncertainty before making population claims.

Two questions for Yaqeen

Yaqeen publishes research, videos and infographics. A useful handoff gives researchers a defensible comparison and product teams a next question they can actually test.

How the original questions connect

Q7 establishes one interview per year and ID for every result. Q1-Q2 describe practice, upbringing and belief. Q3-Q5 branch on current work status: workers for industry and satisfaction, nonworkers for work history. Q6 compares media use across separate waves. These branches need different denominators; a single complete-case filter would discard usable answers.

1 Does afterlife belief vary with reported prayer frequency?

This links Q1 (practice) to Q2 (belief). Within each wave, compare at-least-daily prayer with less frequent prayer. Count only valid yes/no belief answers in the estimate; retain all valid prayer responses for coverage. Weighted yes share = sum of yes weights / sum of valid belief weights in that group.

Wave	Daily or more: valid n / weighted yes	Less than daily: valid n / weighted yes
2014	1,354 / 89.22%	880 / 64.98%
2016	1,576 / 90.36%	995 / 64.35%
2018	1,286 / 89.11%	832 / 68.40%

WTSSPS is the primary weight; the nonresponse-adjusted WTSSNRPS preserves the direction in all three waves. This is an association in the surveyed groups, not a measure of Muslim conviction or a causal effect of prayer. Before a population claim, include survey strata and sampling units. For Yaqeen, ask how its own readers describe belief and practice in their own words.

2 How do web time and daily newspaper reading differ by age?

This extends Q6 using age and weekly non-email web time. Compare both measures among the same respondents, preserving zero hours and excluding skipped items. In 2018:

Age	Paired / known age	Read news daily	Median web hours/week
18-34	376 / 632 (59.5%)	8.0%	14
35-54	479 / 771 (62.1%)	17.1%	10
55+	500 / 938 (53.3%)	30.6%	4

The youngest group reported more web time and less daily newspaper reading than the oldest in each wave. These unweighted pairs exclude Ballot C and people missing either item; 7 interviews also lacked age in 2018. Web time is not internet access, and newspaper reading is not preference for a Yaqeen article. Test a paper and video on the same topic with consenting readers; compare completion and learning separately.

What to operate, and when

Keep the tested local batch now. CI is configured to run dbt on pushes and pull requests and stop publication on failure. At full scale, the AWS sketch uses S3, scheduled ECS tasks and Redshift with separate roles. Add Pew as an independent benchmark and pilot aggregates as a separate product source, never a person-level join. See docs/production.md and docs/analyst-guide.md for assumptions, costs, weights and sources.